

$$I = \int_0^{\infty} \frac{e^{-xt} \sin(xt)}{x^2} dx = ?$$

Let's have  $I(a) = \int_0^{\infty} \frac{e^{-xt} \sin(xt)}{x^2} dx$  for  $a \in \mathbb{R}$ ; thus  $I \equiv I(a)$ .

We use this form in order to cancel  $x^2$  with differentiating 1 with respect to  $a$ . For  $a \in \mathbb{R}$ , let's have  $f_a(x) = \frac{e^{-xt} \sin(xt)}{x^2}$ .

Since  $\lim_{x \rightarrow 0} f_a = a \in \mathbb{R}$  and  $|f_a(x)| = O\left(\frac{1}{x^2}\right)$ , then  $I(a) \in \mathbb{R}$ .

$$\text{Hence } I'(a) = \int_0^{\infty} e^{-xt} \cos(xt) dx$$

$$= \int_0^{\infty} e^{-xt} \operatorname{Re}(e^{iat}) dx$$

$$= \operatorname{Re} \left( \int_0^{\infty} e^{-xt(1-ia)} dx \right) \text{ since } \int_0^{\infty} e^{-xt} \cos(xt) dx \in \mathbb{R}$$

$$= \operatorname{Re} \left( \frac{1}{1-ia} \int_0^{\infty} e^{-x(1-ia)t} dx \right) \text{ where } 1-ia \text{ is the root of}$$

$$= \operatorname{Re} \left( \frac{1}{1-ia} \times \frac{\sqrt{2}}{2} \right)$$

$$\operatorname{Re} \left( \frac{1}{1-ia} \int_0^{\infty} e^{-x(1-ia)t} dx \right) > 0$$

admitting that  $I'(a) > 0$ ,  
since  $e^{-xt} \cos(xt)$  tends  
rapidly towards 0

(we'll justify at the end  
rigorously (2))

$$\text{So } I(a) = \int \operatorname{Re} \left( \frac{\sqrt{2}}{2} \times \frac{1}{1-ia} \right) da$$

$$= \frac{\sqrt{2}}{2} \operatorname{Re} \int \frac{1}{1-ia} da$$

But  $1-ia = (a+bi)$  with  $(a+bi)^2 = 1-ia$   
meaning that  $\begin{cases} a^2 - b^2 = 1 \\ 2ab = -1 \end{cases}$ ,  $2ab = -1$

Since we can write  $\delta_a = (1-i\alpha)^{1/2}$ , we then have

$$I(a) = \frac{1}{2} \operatorname{Re} \int (1-i\alpha)^{-1/2} d\alpha$$

$$= \frac{1}{2} \operatorname{Re} (2 \times i (1-i\alpha)^{1/2} + c), \quad c \in \mathbb{C}$$

But  $I(a) = 0 \Rightarrow \operatorname{Re}(2i+c) = 0 \Rightarrow c = -2i, \quad i \in \mathbb{R}$

$$\text{Hence } I(a) = \frac{\sqrt{2}}{2} \operatorname{Re} (2i(1-i\alpha)^{1/2}) + 0$$

$$= -\sqrt{2} \operatorname{Im} (1-i\alpha)^{1/2}$$

$$= -\sqrt{2} \operatorname{Im}(\delta_a)$$

Since  $a \cdot I(a) \geq 0$  (2): will justify at the end),  $\delta_a$  also verifies  $\operatorname{Im}(\delta_a) < 0 \Rightarrow \beta = -\sqrt{\frac{\sqrt{1+a^2}-1}{2}} \frac{|a|}{a}$ , for  $a \neq 0$

$$\text{Thus } I(a) = \sqrt{\frac{2}{2}(\sqrt{1+a^2}-1)} \frac{|a|}{a} \text{ for } a \neq 0; \quad I(a) = 0$$

Therefore,  $I = \int_0^{\infty} \frac{e^{-st} \sin t}{s^2} ds \quad \text{and } I(a) = \sqrt{\frac{2}{2}(a^2-1)}$

Justification:

(2): let's have  $a \in \mathbb{R}$ . Let's assume without loss of generality that  $a > 0$  (since 1 is odd) and let's show that  $I(a) > 0$

$$I(a) = \int_0^{\infty} \frac{e^{-st} \sin at}{2 \frac{1}{2} \frac{1}{2} \frac{1}{2}} dt$$

$$I_1(a) = \sum_{k \in \mathbb{N}} \underbrace{\int_{\frac{2k\pi}{a}}^{\frac{(2k+1)\pi}{a}} f(u) du}_{I_1(a)} + \underbrace{\int_{\frac{(2k+1)\pi}{a}}^{\frac{2(k+1)\pi}{a}} f(u) du}_{I_2(a)}$$

Let's have  $(t, \pi) \in \left[ \frac{2k\pi}{a}, \frac{(2k+1)\pi}{a} \right] \times \left[ \frac{(2k+1)\pi}{a}, \frac{2(k+1)\pi}{a} \right]$   
for a certain  $k \in \mathbb{N}$ , in such a way that  $\pi =$

$$I_2(a) = \sum_{k \in \mathbb{N}} \int_{\frac{2k\pi}{a}}^{\frac{2(k+1)\pi}{a}} \frac{e^{-t} \sin at}{2t\sqrt{P}} dt \quad \text{let's set } u = t - \frac{\pi}{a}$$

$$I_2(a) = - \sum_{k \in \mathbb{N}} \int_{\frac{2k\pi}{a}}^{\frac{(2k+1)\pi}{a}} \frac{e^{-(u+\frac{\pi}{a})} \sin u}{2(u+\frac{\pi}{a})\sqrt{u+\frac{\pi}{a}}} du$$

We then have  $I_1(a) = \sum_{k \in \mathbb{N}} \int_{\frac{2k\pi}{a}}^{\frac{(2k+1)\pi}{a}} \left( f(t) - f\left(t - \frac{\pi}{a}\right) \right) \sin(at) dt$

where:  $f: t \mapsto \frac{e^{-t}}{2t\sqrt{P}} \xrightarrow{*} \text{over } \mathbb{R}_+^*$

Hence  $I_1(a) > 0$

(2): we demonstrate it similarly by doing the replacements:  $\frac{2k\pi}{a} \rightarrow -\frac{\pi}{2} + 2k\pi$ ,  $\frac{(2k+1)\pi}{a} \rightarrow \frac{\pi}{2} + 2k\pi$ ,

$\frac{2(k+1)\pi}{a} \rightarrow \frac{3\pi}{2} + 2k\pi$ ,  $k \in \mathbb{Z}$ ; yes! there, we can

show that  $\int_{-\pi}^{\pi} e^{ax} \cos(x) dx > 0$  for  $a \in \mathbb{R}$  and

just considerate the case  $a > 0$  since  $a \mapsto \int_{-\pi}^{\pi} e^{ax} \cos(x) dx$  is even; and at the end  $I_1(a) = \int_{-\pi}^{\pi} e^{ax} \cos(x) dx > 0$

Generalisation: Let's have  $I(a) = \int_0^{\infty} f(t) \sin(at) dt$  <sup>strictly</sup>  
 $J(a) = \int_{-\infty}^{\infty} f(t) \cos(at) dt$  where  $a \in \mathbb{R}^*$ ,  $f$  is a <sup>strictly</sup> decreasing  
 function over  $\mathbb{R}_+$  (and is even on  $\mathbb{R}$  for the case of  $J$ ).

Then have:  $* a \cdot I(a) > 0$   
 $* J(a) > 0$

Conclusion: Let's set  $a \in \mathbb{R}$ .

$$I(a) = \int_0^{\infty} \frac{e^{-xt} \sin(at)}{x} dx = \sqrt{\frac{\pi}{2}} (\sqrt{1+a^2} - 1) \times \frac{a}{|a|}$$

~~$$= \sqrt{\frac{\pi}{2}}$$~~

$$= a \sqrt{\frac{\pi}{2} \times \frac{1}{1+\sqrt{1+a^2}}}$$

And  $I = \int_0^{\infty} \frac{e^{-xt} \sin xt}{x} dx = \sqrt{\frac{\pi}{2(1+\sqrt{2})}}$